

Laurentian Hickey
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Garson ON,
P3L 1H3

To whom it may concern,

I am writing to express my enthusiastic interest in the various positions at your company. With a robust background in software development, coupled with extensive experience across a diverse technology stack, I am confident in my ability to contribute significantly to your team.

The reason for this confidence is not due to my wide breadth of knowledge in different languages but due to my strong and proven ability to pick up and learn new ones very quickly under tight deadlines.

In my current roles, I continually find myself learning and acquiring new skills that push me to adapt to the new challenges I face to make the final products as polished as possible.

What I've learned is of the utmost importance, when working on any of my projects is first and foremost fully understanding the requirements and learning as much as possible from the associated documentation. This has led me through much smoother and faster development cycles and in many cases has required less research for related projects over time.

The nature of my employments has afforded me the privilege of also being a software developer that in many cases, can provide unique solutions to intricate problems that would otherwise have not been considered by those who may have received a more traditional career path.

I thank you for considering my application. I have attached my resume for your review and welcome the opportunity to discuss how my skills and experience can benefit your team and look forward to growing with your organization.

Lauren H.



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LAURENTIAN HICKEY

HEAD OF GAMIFICATION

CONTACT

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EDUCATION

2017-2021
LAURENTIAN UNIVERSITY

- Bachelor of **Computer Science** with a specialisation in **Game Design**

SKILLS

- Project Management
- Waterfall/Agile Methodologies
- Interdisciplinary Collaboration
- Time Management & Prioritization
- Leadership & Mentorship
- Effective Communication
- Problem Solving & Debugging

LANGUAGES

- GDScript
- Kotlin
- Python
- Java
- C#
- MySQL
- Git
- XML
- Web fundamentals (HTML, CSS, JavaScript, TypeScript)
- C++
- Extremely adaptive to new languages and concepts

PROFILE SUMMARY

Full-stack Software Developer working primarily with 2D/3D game engines and Native android development with both Java/Kotlin. Embedded systems programming and integration layers, Firebase queries, MySQL, web as well as Godot integrations for dental software solutions.

WORK EXPERIENCE

IRegained Inc. 2022-PRESENT
Software Developer

IRegained Inc. 2024-PRESENT
Head of Gamification & Data Analysis

- Developed** and **integrated** 20+ 2D **Unity** games into an Android application, later migrating to **Godot** to create a new series of 2D and 3D games with Android integration
- Designed** and implemented a **facial recognition** system leveraging **Machine Learning Kit**, and developed systems to interpret force values from CSV data for mechanical force applications
- Contributed to core **Android** development, including native activities, **Python** script integration for data charting, and **Bluetooth** functionality
- Implemented robust security measures to protect proprietary software and managed software documentation, testing, review, and deployment processes
- Created **custom** controls for **user-tailored** game activities and designed unique, comparable scoring systems across games
- Managed** device security and deployment for Android products, including Mobile Device Management (MDM) integration
- Developed a **sandbox** environment to enable and streamline onboarding game development **mentorship**
- Engineered** data conversion scripts to transform raw .csv data into proprietary .json formats for dynamic activity use
- Designed** a milestone progression system that outlines clear goals and tracks user improvement
- Developed **intuitive design** choices to effectively convey **real-world**, relevant progress to individual users
- Designed and facilitate the use and compliance of an **ISO 13485** QMS System

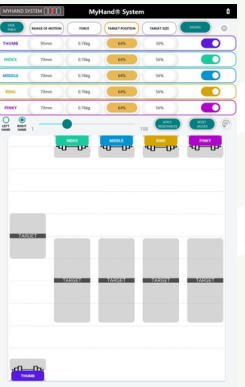
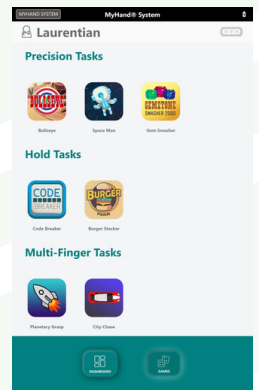
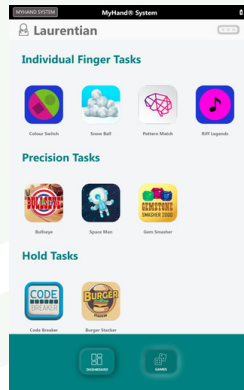
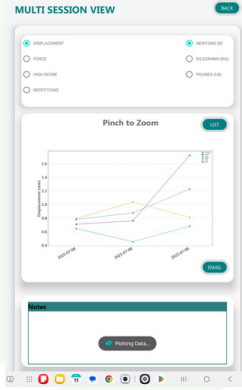
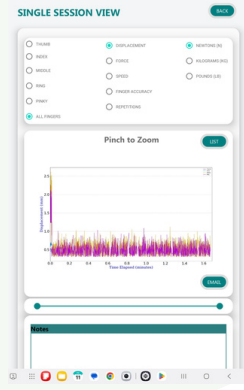
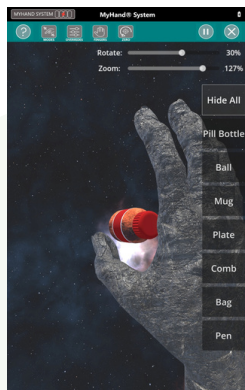
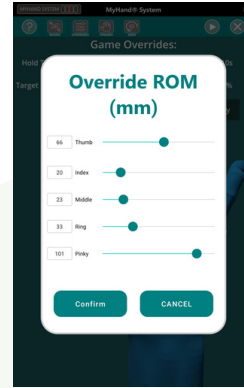
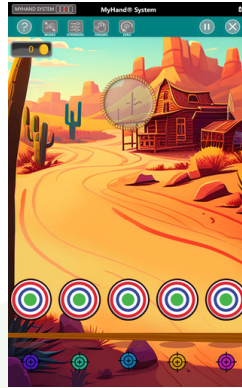
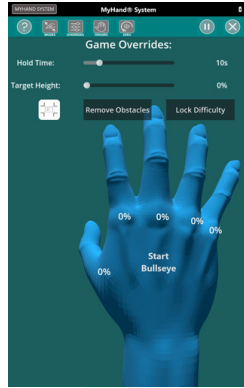
AlgoLibre 2025-PRESENT
Software Developer

- Godot front end and back end development with database integrations for dental software solutions with networking components required for paging services

PORTFOLIO HIGHLIGHTS

PERSONAL: [HTTPS://LITTLEMANGIT.ITCH.IO/](https://littlemangit.itch.io/)

PROFESSIONAL: SEE BELOW



PATIENT INFO

CHOOSE AFFECTED HAND

CHOOSE WEIGHT UNIT

ENTER FINGER DATA

	AFFECTED		UNAFFECTED	
	FORCE (KG)	RANGE OF MOTION (DEG)	FORCE (KG)	RANGE OF MOTION (DEG)
THUMB	0.17	140.0	1.00	100.0
INDEX	0.2	200.0	1.0	200.0
MIDDLE	0.09	250.0	1.0	300.0
RING	0.2	310.0	1.0	400.0
PINKY	0.17	300.0	1.0	500.0

SELECT PROTOCOL

Device Height

Finger Position

```
fun readForceTable(context: Context, filename: String): Array<DoubleArray> {  
    //check if unique table exists  
    val file = File(pathname: "${Environment.getExternalStoragePublicDirectory(Environment.DIRECTORY_DOWNLOADS)}" +  
        "/MyHand8/Content/${(packageName.split(".").last()).split(".").last()}_${filename}")  
    val inputStream: InputStream?  
    if (file.exists()) inputStream = FileInputFileStream(file)  
    else { //use generic if nothing else  
        val assetManager: AssetManager = context.assets  
        inputStream = assetManager.open(filename)  
    }  
    val rowsList = mutableListOf<List<Double>>()  
    BufferedReader(BufferedReader(inputStream)).use { reader ->  
        reader.readLine() //skip column headers  
        reader.readLine() //skip 0 displacement row  
        reader.forEachLine { line ->  
            val row = line.split(",").map { it.trim().toDouble() }  
            rowsList.add(row)  
        }  
    }  
    val rowsArray: Array<DoubleArray> = Array(rowsList.size) { DoubleArray(rowsList[i].size) }  
    for (i in rowsList.indices) {  
        rowsArray[i] = rowsList[i].toDoubleArray()  
    }  
    return rowsArray  
}
```

```
func _get_smoothness(values: Array<float>, sample: int):  
    readingsCount += 1  
    var averages: Array<int> = [0,0,0,0,0]  
    for i in smoothnesses.size():  
        if smoothnesses[i].size() < sample:  
            smoothnesses[i].append(values[i])  
        else:  
            smoothnesses[i].pop_front()  
            smoothnesses[i].push_back(values[i])  
    return max difference per finger  
    averages[i] = int(((smoothnesses[i].max() - smoothnesses[i].min()) * 100))  
    if all values equal 0, fingers not moving  
    if smoothnesses[i].all(func(element): return element == 0):  
        Global.stabilized[i] = true  
    else:  
        Global.stabilized[i] = false  
    Global.smoothness = averages.duplicate()  
    return a singular average for all active fingers only  
    var average = averages.reduce(func(total, value): return total + value, 0) / Global.fingerCount  
    only display change after x samples parsed  
    if readingsCount > 5:  
        readingsCount = 0
```



```
if finger == 0 and thumb.shape[0] > 0:  
    flex = thumbFlex.to_numpy()  
    ext = thumbExt.to_numpy()  
    resRows = thumb["resistance"].to_numpy().astype(int)  
    disRows = thumb["displacement"].to_numpy().astype(int)  
    direction = thumb["actuatorDirection"].to_numpy()  
    flexion_mask = direction == "FLEXION" #boolean list to know if entry is flexion or extension  
    force_values = np.zeros(resRows.shape[0]) #pre-allocate space  
    #return all needed flexion or extension entries at given res indices  
    force_values[flexion_mask] = flex[disRows[flexion_mask] - 2, resRows[flexion_mask]]  
    force_values[-flexion_mask] = ext[disRows[-flexion_mask] - 2, resRows[-flexion_mask]]  
    if units == "LB":  
        force_values /= 4.4482216153  
    elif units == "KG":  
        force_values /= 9.80665  
    thumbForces = np.round(force_values, 5)  
    subplot.plot(stringToDate(thumb) - stringToDate(thumb)[0] / 40, thumbForces,  
        color='g', ls = '-' if fileIndex == 0 else '--')
```